Ex. No: 11 Memory Allocation Methods

PROGRAM A: FIRST FIT

C PROGRAM

```
#include <stdio.h>
int main()
{
int blockSize[20], processSize[20];
int allocation[20];
int nb, np, i, j;
printf("Enter Number of Blocks: ");
scanf("%d",&nb);
printf("Enter Number of Processes: ");
scanf("%d",&np);
printf("Enter Block Sizes:\n");
for(i=0;i<nb;i++)
scanf("%d",&blockSize[i]);
printf("Enter Process Sizes:\n");
for(i=0;i<np;i++)
scanf("%d",&processSize[i]);
for(i=0;i<np;i++)
allocation[i]=-1;
for(i=0;i<np;i++)
{
for(j=0;j<nb;j++)
{
if(blockSize[j] >= processSize[i])
{
allocation[i]=j;
blockSize[j]-=processSize[i];
break;
}
```

```
}
}
printf("\nProcess No\tProcess Size\tBlock No\n");
for(i=0;i<np;i++)
{
printf("%d\t\t%d\t\t",i+1,processSize[i]);
if(allocation[i]!=-1)
printf("%d\n",allocation[i]+1);
else
printf("Not Allocated\n");
}
return 0;
}
```

Output:

SHELL SCRIPT

```
#!/bin/bash
echo "First Fit Memory Allocation Demonstration"
blocks=(100 500 200 300 600)
processes=(212 417 112 426)
echo "Memory Blocks: ${blocks[@]}"
echo "Processes: ${processes[@]}"
echo "Allocation Performed Using First Fit"
```

Output:

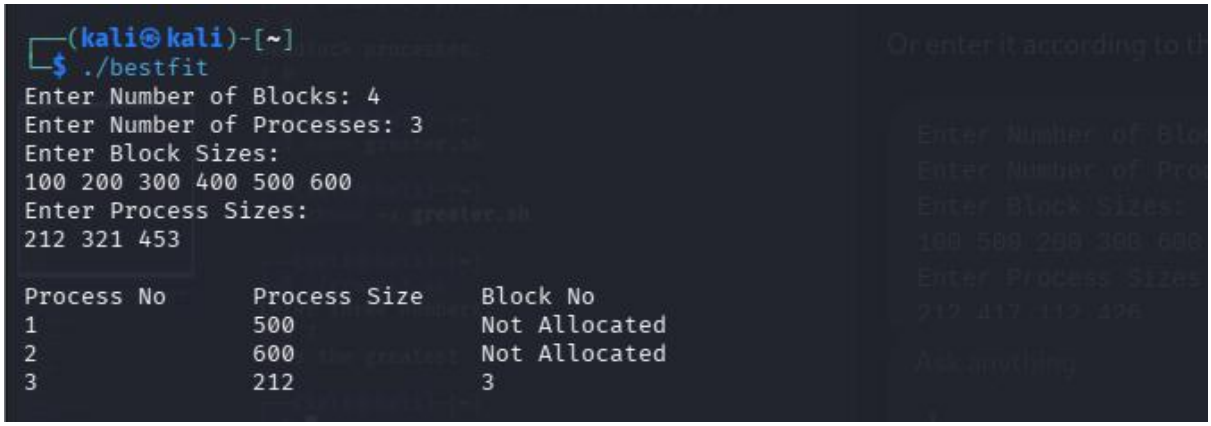
PROGRAM B: BEST FIT

C PROGRAM

```
#include <stdio.h>
int main()
{
int blockSize[20], processSize[20];
int allocation[20];
int nb,np,i,j,bestIdx;
printf("Enter Number of Blocks: ");
scanf("%d",&nb);
printf("Enter Number of Processes: ");
scanf("%d",&np);
printf("Enter Block Sizes:\n");
for(i=0;i<nb;i++)
scanf("%d",&blockSize[i]);
printf("Enter Process Sizes:\n");
for(i=0;i<np;i++)
scanf("%d",&processSize[i]);
for(i=0;i<np;i++)
allocation[i]=-1;
for(i=0;i<np;i++)
{
bestIdx=-1;
for(j=0;j<nb;j++)
```

```
{
if(blockSize[j] >= processSize[i])
{
if(bestIdx==-1 || blockSize[j] < blockSize[bestIdx])
bestIdx=j;
}
}if(bestIdx!=-1)
{
allocation[i]=bestIdx;
blockSize[bestIdx]-=processSize[i];
}
}
printf("\nProcess No\tProcess Size\tBlock No\n");
for(i=0;i<np;i++)
{
printf("%d\t\t%d\t\t",i+1,processSize[i]);
if(allocation[i]!=-1)
printf("%d\n",allocation[i]+1);
else
printf("Not Allocated\n");
}
return 0;
}
```

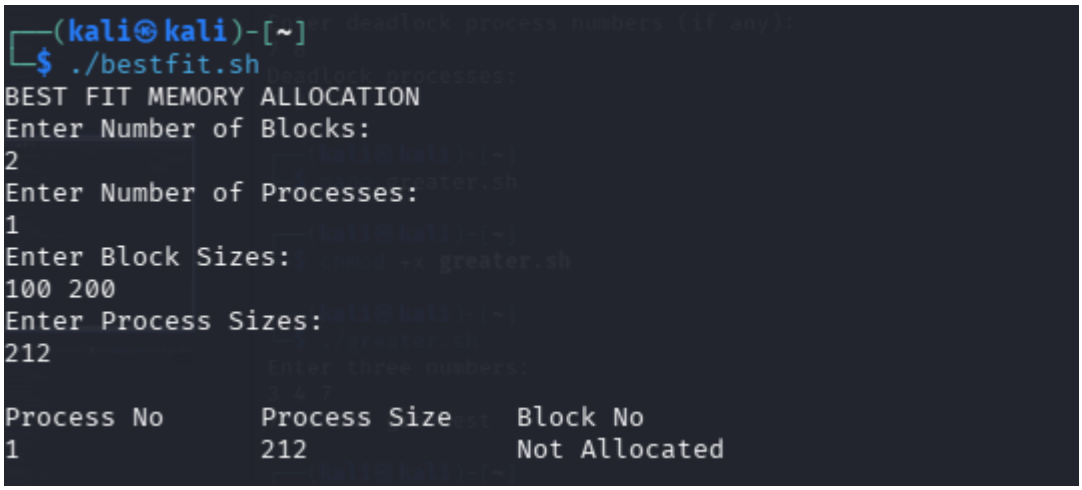
Output:



SHELL SCRIPT

```
#!/bin/bash
echo "Best Fit Memory Allocation Demonstration"
blocks=(100 500 200 300 600)
processes=(212 417 112 426)
echo "Memory Blocks: ${blocks[@]}"
echo "Processes: ${processes[@]}"
echo "Allocation Performed Using Best Fit"
```

Output:



PROGRAM C: WORST FIT

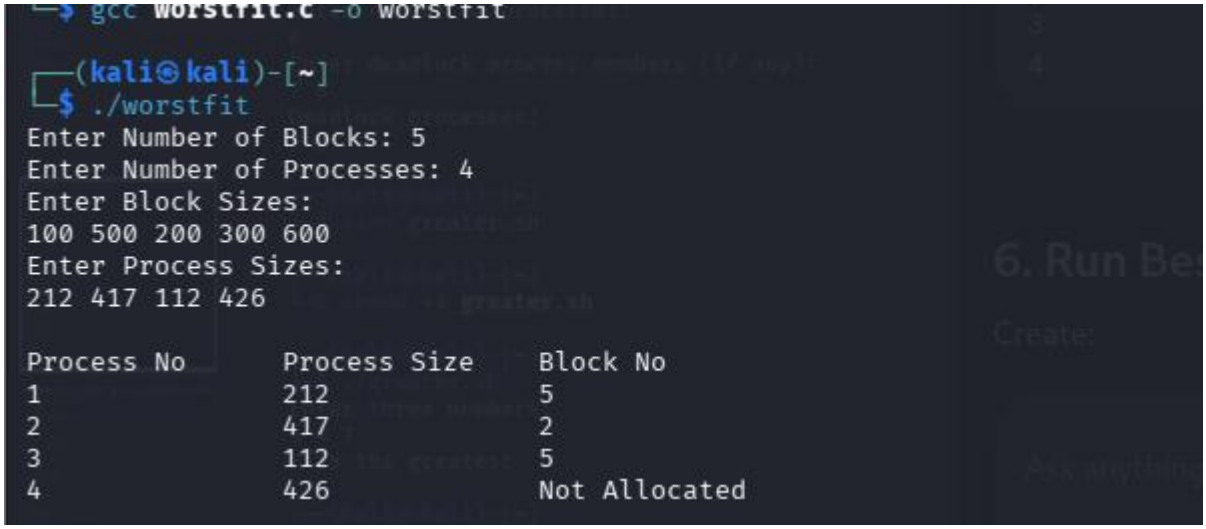
C PROGRAM

```
#include <stdio.h>
int main()
{
int blockSize[20], processSize[20];
int allocation[20];
int nb,np,i,j,worstIdx;
printf("Enter Number of Blocks: ");
scanf("%d",&nb);
printf("Enter Number of Processes: ");
scanf("%d",&np);
```

```
printf("Enter Block Sizes:\n");
for(i=0;i<nb;i++)
scanf("%d",&blockSize[i]);
printf("Enter Process Sizes:\n");
for(i=0;i<np;i++)
scanf("%d",&processSize[i]);
for(i=0;i<np;i++)
allocation[i]=-1;
for(i=0;i<np;i++)
{
worstIdx=-1;
for(j=0;j<nb;j++)
{
if(blockSize[j] >= processSize[i])
{
if(worstIdx==-1 || blockSize[j] > blockSize[worstIdx])
worstIdx=j;
}
}
if(worstIdx!=-1)
{
allocation[i]=worstIdx;
blockSize[worstIdx]-=processSize[i];
}
}
printf("\nProcess No\tProcess Size\tBlock No\n");
for(i=0;i<np;i++)
{
printf("%d\t\t%d\t\t",i+1,processSize[i]);
if(allocation[i]!=-1)
printf("%d\n",allocation[i]+1);
```

```
else
printf("Not Allocated\n");
}
return 0;
}
```

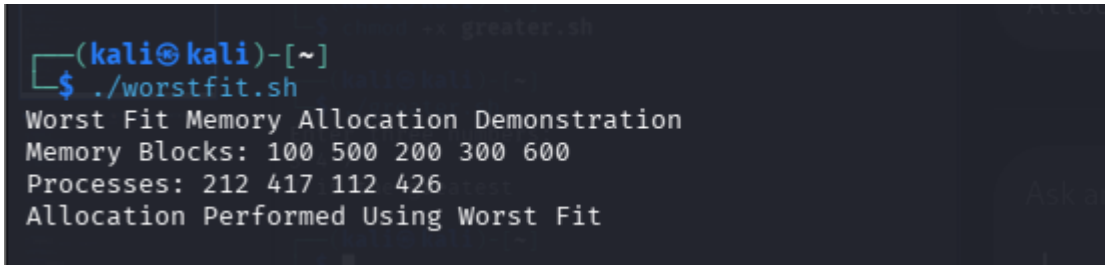
Output:



SHELL SCRIPT

```
#!/bin/bash
echo "Worst Fit Memory Allocation Demonstration"
blocks=(100 500 200 300 600)
processes=(212 417 112 426)
echo "Memory Blocks: ${blocks[@]}"
echo "Processes: ${processes[@]}"
echo "Allocation Performed Using Worst Fit"
```

Output:


(kali㉿kali)-[~]
$ ./worstfit.sh
Worst Fit Memory Allocation Demonstration
Memory Blocks: 100 500 200 300 600
Processes: 212 417 112 426
Allocation Performed Using Worst Fit